



ASSIGNMENT 3:
for
GOOGLE STOCK PRICE

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TABLE OF CONTENTS

1.0 INTRODUCTION	2
1.1. Background Of The Company	2
1.2. Issue and Motivation	2
1.3. Brief Explanation of Methods Used	3
2.0 RESEARCH OBJECTIVES	4
3.0 DATA DESCRIPTION	4
3.1. Analysis of the Time Series Components	5
3.1.1. Trend Component	5
3.1.2. Cyclical Component	6
3.1.3. Seasonal Component	6
3.1.4. Irregular Component	7
4.0 METHODOLOGY	7
4.1. Univariate Modelling Techniques	7
4.2. ARIMA / SARIMA Modelling - Box-Jenkins Methodology	10
5.0 ANALYSIS AND RESULT	17
5.1. Univariate Modelling Technique	17
5.2. ARIMA Models	31
6.0 CONCLUSION	47
7.0 REFERENCE	48
8.0 APPENDICES	50

1.0 INTRODUCTION

1.1. Background Of The Company

Alphabet Inc. is the parent company of Google among other entities. Shares of their Class. A substantial portion of Google Services revenue within Alphabet comes from advertising on Google Search, YouTube, and the Google Network. Alphabet owns Google Cloud and is seemingly invested in their offerings and services related to their interests in artificial intelligence, data analytics, cybersecurity, business applications, and consumer devices (Alphabet Inc., 2025). Due to Google's high shares and significant market presence, Google is an excellent candidate for time series analysis and stock performance forecasting.

The data for this analysis was downloaded from the Google 10 Years Stock Price Dataset on Kaggle (Jillani Soft Tech, n.d.). The file GoogleStock_Dataset_V2.csv was created from 2,775 daily records between January 2, 2013, and January 10, 2024. The daily closing prices represent a substantial increase over the given time period, however, the dataset captures many daily price declines, price fluctuations, daily price peaks and price valleys, and a principal major decline in the year 2022 followed by a notable increase in the year 2023. This dataset provides a long-term price trend with short-term price fluctuations caused by the market and company-specific events.

1.2. Issue and Motivation

To a degree, the stock price and trade volume data help gauge investor sentiment and aid understanding of the trader's stock selling/liquidating and risk hedging strategies and the resulting price volatility. However, predicting price changes is a challenging endeavor. Stock interest fluctuates with the arrival of new information: changes in the company's earnings, new economic reports, changes in general stock market sentiment positive or negative as well as price changes, inventions, technological changes, and news changes. According to the Efficient Market Hypothesis, stock prices rapidly adjust to reflect news, thus rendering all prior price information nearly useless for predicting future prices (Fama, 1970).

This research aims to determine the best forecasting method to predict future daily closing prices of Google, Inc. The existing data will be analyzed to identify optimal forecasting methods. Different forecasting methods will be implemented and the performance of each method will be assessed on data not used to implement the method. In line with the industry recommendations, 30% of the data will be set aside for evaluation. The primary focus of the project will be the analysis of the forecasting methods assessing both the Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE).

1.3. Brief Explanation of Methods Used

Applying three different exponential smoothing methods will be the first approach. These methods are Single Exponential Smoothing (SES), Holt's Method, and Adaptive Response Rate Exponential Smoothing (ARRES). Holt's Method, SES, and ARRES are often termed as level and trend smoothing (Holt, 2004; Hyndman et al., 2008). Holt's Method is the only one of the three that incorporates a trend. SES is a basic level model. ARRES is the most flexible of the three as it adapts the smoothing value when the time series is affected by a structural change. The second approach uses the Box–Jenkins methodology. This includes the development of ARIMA models and also the SARIMA models when seasonality is present. The process of modeling (Box et al., 2015) includes the phases of identifying, estimating, validating, and forecasting. Candidate models will be checked using the AIC, BIC, and the Ljung–Box test. The best univariate model, based on the same hold-out data and error measures (Ljung & Box, 1978) will be the benchmark for the SARIMA and ARIMA models.

2.0 RESEARCH OBJECTIVES

The objectives of this study are:

- To study the pattern and behavior of the daily Google closing price series, including its trend, cyclical, seasonal, and irregular components.
- To determine the most suitable forecasting model for the series by comparing univariate modeling techniques with the Box-Jenkins ARIMA methodology.
- To forecast the future daily closing price of Google using the best-fitted model and to evaluate its accuracy on a holdout sample.

3.0 DATA DESCRIPTION

The dataset used in this study is the *Google 10 Years Stock Price Dataset*, available on Kaggle (Jillani Soft Tech, n.d.). It consists of 2,775 records of daily closing prices of Alphabet Inc. Class A, traded as Google. The data spans the period from 2 January 2013 to 10 January 2024. The dataset contains only records of the trading days. The data available corresponds to the days in which the NASDAQ stock market was open.

This study examines the end-of-day final trade prices for Google shares. The final trade price of the day is referred to as the daily trade price. Daily movements of the closing price are shown in Figure 3.1. Daily trade dates are shown on the x-axis, while the daily closing trade price in dollars is shown on the y-axis.



Figure 3.1 Daily closing price of Google from 2 January 2013 to 10 January 2024, showing the estimation and hold-out samples.

As shown in Figure 3.1, Google share prices rose steadily throughout the period of study. Prices in 2013 started at about USD 18, and ended at about USD 144 in January 2024. It is important to note that the price journey was segmented by periods comprising sharp price movements in the upward and downward directions, and periods of price consolidations. Between 2013 and 2019, the price movement was generally in the positive direction, but slower. This was followed by a period of price movement in the positive direction that was much faster in 2020 and 2021. Prices in 2022 recorded a sharp downward movement. Prices generally started to move in the upward direction during the 2023 and this was referred to as a price recovery movement.

3.1. Analysis of the Time Series Components

3.1.1. Trend Component

The time series describes an evident long term increase. A time series trend describes a long term increase or decrease movement. In the examined dataset, the long term the closing price has an upward movement from around USD18 in 2013 to around USD143 in 2024.

The closing price's long term upward movement describes a long term non-stationary movement. A movement is described as stationary when it is centered around a mean. Throughout the examined period the price of Google has an upward movement, therefore, a univariate method with a trend, for example, Holt's Method, would be appropriate. It also shows that an ARIMA model would be appropriate after a differencing is applied.

3.1.2. Cyclical Component

A cyclical component can be identified as a medium-term series characterized by the absence of regularity as well as precision in upward and downward movements. Several such series can be seen in Figure 3.1. These movements especially include the significant increase from 2020 to 2021, the rapid decline in 2022, as well as the increase that occurred in 2023.

Due to the extended duration of cyclical-like movements as compared to the usual fluctuations of a series, these movements can be analyzed with greater precision. However, greater irregularity of movements precludes classifying these movements as fixed cycles. From the graph, it is not possible to identify the movements as being caused by a particular business or economic cycle. These movements are better described as medium-term fluctuations of a series resulting from prevailing market conditions, investors' expectations, and the performance of the business.

3.1.3. Seasonal Component

A movement that occurs at regular, predictable intervals is known as a pattern that involves seasonality. In reference to Figure 3.1, there are no recognizable patterns therefore no seasonality of daily GOOGLE closing prices can be established. No movements are seen to occur at regular or predictable intervals.

Nonetheless, the absence of a definitive trend of a time series in a given graph is only the preliminary indication of the absence of seasonality patterns, and is not to be taken as the absence of seasonality patterns in time series. The

autocorrelation function will be analyzed in regard to significant relations at potential seasonal lags. In the absence of relevant seasonal patterns, a non-seasonal ARIMA will be preferred to a SARIMA, and the Holt-Winters and other seasonal methods will not be applicable.

3.1.4. Irregular Component

The irregular component covers all alterations remaining after accounting for trend, cyclical and seasonal behaviors, and cannot be predicted. In relation to text Figure 3.1, irregular components are day-to-day variations that elapse the general trend of a data series. In stock price transactional behavior, after price trends have been established, price movements can trend upwards one day and downwards the following day.

Unpredictable factors are released firm-related information due to firm announcements, changes in the state of the economy, alterations to interest rates, fluctuations in the overall market and/or in other equities, and shifts in investor behavior. The modeling of irregular components is the primary reason daily price fluctuations are so challenging to predict. A successful model is built on the structure of historical data and accounts for daily price fluctuations.

4.0 METHODOLOGY

4.1. Univariate Modelling Techniques

Univariate forecasting models use the historical values of a single variable to predict its future values. Although these models can identify and forecast patterns within the series, they do not explain the external causes of changes in the variable. Three exponential smoothing techniques were selected for the univariate analysis: Single Exponential Smoothing (SES), Holt's method, and Adaptive Response Rate Exponential Smoothing (ARRES). SES is a simple level smoothing technique. Holt's method adds a linear trend, and ARRES permits time-varying smoothing constants. Thereafter, the three techniques were

compared using mean squared error (MSE) and mean absolute percentage error (MAPE) for both the estimation and hold-out samples.

For SES and Holt's method, the smoothing parameters corresponded to the minima of the estimation sample losses. For ARRES, the initial value of the smoothing constant was specified to be $\alpha_1 = 0.6$, and the error-smoothing constant to be $\beta = 0.2$. The value of α_t was updated recursively over time. The resulting forecasts were compared using MSE and MAPE to determine the most appropriate univariate forecasting method.

1. Single Exponential Smoothing (SES)

SES generates fitted values and forecasts using a single smoothing constant α :

$$F_{t+1} = \alpha y_t + (1 - \alpha)F_t, \quad 0 \leq \alpha \leq 1$$

It was selected as a baseline model for its benefits over moving averages as well as its simplicity. It gives most of its weight to the more recent observations and stores less data. Because of this, previous observations from many different time periods do not need to be held. The method assigns a start-up value which can be either the first observation in the series, or the average of the first five observations. It then computes the value of the smoothing constant, α , that produces the least Mean Squared Error (MSE) for the model. The main disadvantage of the method, which is also the main reason why other models have been developed, is that it is a one-step-ahead forecasting model, and it does not account for trends.

2. Holt's Method

To address the trend limitation of SES, Holt's Method was employed. It smooths the level and trend components separately using two smoothing constants:

$$\textbf{Level: } L_t = \alpha y_t + (1 - \alpha)(L_{t-1} + T_{t-1})$$

$$\textbf{Trend: } T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$$

$$\textbf{Forecast: } F_{t+m} = L_t + T_t \times m$$

with $0 \leq \alpha \leq 1$ and $0 \leq \beta \leq 1$. Holt's Method was selected because the Google daily closing-price series exhibits a clear long-term trend, it provides flexibility in selecting the rates at which the level and trend components are tracked (through separate constants), and unlike SES it can forecast m -steps ahead for $m = 1, 2, \dots, n$.

3. Adaptive Response Rate Exponential Smoothing (ARRES)

Within SES, α is presumed time invariant. However, an event may cause the behavior of the series to change either temporarily or permanently, and in that case, α may not be invariant. For this reason, ARRES was developed. It is designed to adjust to the changes in the series and to allow changes to the smoothing constant:

$$F_{t+1} = \alpha_t y_t + (1 - \alpha_t) F_t,$$

$$\alpha_t = \left| \frac{E_t}{AE_t} \right|$$

$$E_t = \beta e_t + (1 - \beta) E_{t-1},$$

$$AE_t = \beta |e_t| + (1 - \beta) AE_{t-1}$$

where $e_t = y_t - F_t$, $0 < \beta < 1$, E_t is the smoothed average error and AE_t is the smoothed absolute error. The adaptive ratio α_t rises when forecast errors are consistently in one direction (the series has shifted) and falls when errors are random, letting the model respond quickly to genuine changes while remaining stable otherwise.

In SES and Holt's method, the smoothing parameters were derived from the objective of minimising the sum of squared errors (SSE) over the estimation sample. With a finite sample size, minimising SSE is the same as minimising MSE. For ARRES, an initial smoothing constant of α_1 was set to 0.6, and the error-smoothing constant was set to β of 0.2 and the α_t was updated recursively at each time period.

4.2. ARIMA / SARIMA Modelling - Box-Jenkins Methodology

The Box-Jenkins method involves a rigorous process for developing time series forecasting models based on AutoRegressive Integrated Moving Average (ARIMA) through identification, estimation and evaluation. The process involves four iterative stages:

Stage 1: Model Identification

This stage aims for stationarity and determining the preliminary model orders. The first step involves a time series plot to identify any visible trends and seasonality. To identify stationarity, the Augmented Dickey-Fuller (ADF) test is employed. If the test provides a p-value less than the significance level, typically set at (0.05), the null hypothesis is rejected and can be concluded that the series is stationary.

The ADF test formally assesses whether the series is stationary:

H_0 : The series has a unit root (the series is non-stationary)

H_a : The series has no unit root (the series is stationary)

The Phillips–Perron (PP) and KPSS tests were used to determine whether the Google daily closing-price series was stationary. Both tests were conducted at the 5% significance level

Table 4.1: PP test and KPSS test hypothesis

Test	Null hypothesis, (H_0)	Decision indicating stationarity
PP test	The series contains a unit root and is non-stationary	Reject (H_0) when ($p < 0.05$).
KPSS test	The series is stationary	Fail to reject (H_0) when ($p > 0.05$)

The PP and KPSS tests have opposite null hypotheses. Therefore, stationarity is strongly supported when the PP test produces a p-value below 0.05, while the KPSS test produces a p-value above 0.05. The tests were first applied to the original series and then repeated after first-order differencing.

If the data is found to be non-stationary, differencing techniques are applied:

- If the series naturally fluctuates around a constant value, an **ARMA(p,q)** model is used.
- If a trend is present, first-order differencing ($\Delta y_t = y_t - y_{t-1}$) is applied, leading to an **ARIMA(p,d,q)** model where d indicates the order of differencing.
- If a seasonal component is present, seasonal differencing ($z_t = y_t - y_{t-s}$, where s is the seasonal length) is performed. If the resulting series remains non-stationary, a further non-seasonal difference ($w_t = z_t - z_{t-1}$) is applied, yielding a **SARIMA(p,d,q)(P,D,Q)s** model.

Once stationarity is established, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots are examined. The ACF is utilized to suggest the Moving Average (MA) and Seasonal Moving Average (SMA) orders, while the PACF suggests the AutoRegressive (AR) and Seasonal AutoRegressive (SAR) orders.

The standard error of the autocorrelation at lag k is:

$$SE_{ACF(k)} = \frac{1}{\sqrt{T}}$$

where T is the number of observations in the series. The 95% confidence interval for each autocorrelation r_k is:

$$r_k \pm 1.96 \cdot \frac{1}{\sqrt{T}}$$

Spikes that fall outside this interval are considered statistically significant at the 5% level, and are used to determine the model orders.

Model Identification Guide (ACF & PACF patterns):

- Model : ARMA(p, q)
- ACF Pattern : k significant spikes
- PACF Pattern : k significant spikes
- Special Condition : If ACF shows one significant spike at lag 1 AND PACF shows one significant spike at lag 1, the suggested model is ARMA(1,1)

Stage 2: Model Estimation

After the preliminary models and their neighboring competition are established, the candidate models' parameters are assessed. Parameter assessment methods, such as Ordinary Least Squares (OLS) and Maximum Likelihood, involve a computation where the sum of squared residuals is minimized. During the parameter assessment, it is important to avoid parameter redundancy, as various ARMA specifications may be mathematically equivalent.

The Backshift (Lag) Operator :

The backshift operator B is a mathematical convenience used to express ARIMA and SARIMA models compactly. It is defined as:

$$By_t = y_{t-1}$$

More generally:

$$B^k y_t = y_{t-k}$$

Using this notation, the differencing operations can be written as:

- Non-seasonal differencing

$$\Delta y_t = y_t - y_{t-1} = (1 - B)y_t$$

- Seasonal differencing (of order s)

$$z_t = y_t - y_{t-s} = (1 - B_s)y_t$$

- Combined seasonal and non-seasonal differencing

$$w_t = z_t - z_{t-1} = (1 - B)(1 - B_s)y_t$$

General Model Expressions Using the Backshift Operator

- ARIMA(p, d, q):

$$\phi(B)(1 - B)^d y_t = \theta(B)\varepsilon_t$$

where:

- $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$ is the AR polynomial
- $\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$ is the MA polynomial
- $(1 - B)^d$ Represents d orders of non-seasonal differencing
- ε_t is white noise
- SARIMA(p, d, q)(P, D, Q)s:

$$\Phi(B^s)\phi(B)(1 - B^s)^D (1 - B)^d y_t = \Theta(B^s)\theta(B)\varepsilon_t$$

- $\Phi(B^s) = 1 - \Phi_1 B^s - \dots - \Phi_P B^{Ps}$ is the seasonal AR polynomial
- $\Theta(B^s) = 1 + \Theta_1 B^s + \dots + \Theta_Q B^{Qs}$ is the seasonal MA polynomial
- $(1 - B^s)^D$ represents D orders of seasonal differencing
- s is the seasonal period

Stage 3: Model Validation

The estimated models undergo diagnostic checking to ensure they are well-specified. To check if the residuals behave as “white noise,” we check if they are normally, independently, and randomly distributed with a mean of zero and a constant variance. In this case, “white noise” can be thought of as the ideal case.

The Box-Pierce Q statistic or the Ljung-Box statistic will formalize the above. The null hypothesis is that the residuals are white noise. The alternative hypothesis is that the residuals are not white noise. The model is well-specified if we fail to reject the null hypothesis, with a significance level greater than $p = 0.05$.

This assumption is formally tested using the Box-Pierce Q statistic or Ljung-Box statistic with the following hypotheses:

H_0 : The residuals are white noise (model is adequate)

H_a : The residuals are not white noise (model is inadequate)

The number of lags used in the test depends on the sample size and data type:

- If the sample size is n , the lag is $\sqrt{n}$
 - Example: if $n = 100$, the $lag = \sqrt{100} = 10$
- For a seasonal period, the lag is $2m$, where m is the length of the seasonal period
 - Example: for monthly data, $m = 12$, hence $lag = 12 \times 2 = 24$
- When testing residuals from an ARIMA model, the degrees of freedom must be adjusted for the number of estimated parameters

Among the thoroughly defined models, the last model is chosen by reducing Akaike’s Information Criterion (AIC), AIC’s correction (AICc), and Bayesian’s

Information Criterion (BIC). These criteria measure and compare the statistical model's quality, 'AIC'; less is more, the demand is for the simplest model.

Akaike's Information Criterion (AIC):

AIC is a statistical measure used to estimate the relative quality of competing models for a given dataset. It is written as:

$$AIC = -2\log(L) + 2(p + q + k + 1)$$

where:

- L is the likelihood of the data
- $k = 1$ if $\phi_0 \neq 0$, and $k = 0$ if $\phi_0 = 0$
- Lowest AIC values indicate a better fit of the model to the data

For ARIMA models, the Corrected AIC (AICc) is used to penalize model complexity more appropriately for smaller sample sizes:

$$AICc = AIC + \frac{2(p+q+k+1)(p+q+k+2)}{T-p-q-k-2}$$

AICc penalizes model complexity by adding twice the number of parameters, discouraging overly complex models.

Bayesian Information Criterion (BIC)

BIC is similar to AIC but imposes a stronger penalty for model complexity. It is written as:

$$BIC = AIC + [\log(T) - 2](p + q + k + 1)$$

Like AIC , lower BIC values indicate a better fit of the model to the data

BIC penalizes complexity more heavily than AIC because it includes a term that grows with the sample size T , making it more conservative in selecting parsimonious models

Stage 4: Model Application

The finalized validated model is employed for generating point forecasts. The ARIMA equation is modified to reflect the dependent variable, y_t , on the left side of the equation.

The general ARIMA(p, d, q) model expressed in terms of y_t is:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}$$

where:

- c is a constant
- $\phi_1, \phi_2, \dots, \phi_p$ are the AR parameters
- $\theta_1, \theta_2, \dots, \theta_q$ are the MA parameters
- ε_t is the white noise error term at time t

To compute an h -step-ahead forecast, we substitute t with $T + h$. On the equation's right side, the future values are recursively replaced by the respective forecasts, the future errors are replaced by zero, and the past errors are replaced by the respective residuals. This recursive technique extends the point forecasts to any desired number of future periods.

To compute an h -step-ahead forecast, we substitute t with $T + h$:

$$\hat{y}_{T+h|T} = c + \phi_1 \hat{y}_{T+h-1|T} + \dots + \phi_p \hat{y}_{T+h-p|T} - \theta_1 \varepsilon_{T+h-1} - \dots - \theta_q \varepsilon_{T+h-q}$$

The following substitution rules apply recursively on the right-hand side:

Table 4.2: Term and condition of Step ahead Forecast

Term	Condition	Replacement
y_{T+h-i}	$h - i > 0$ (future value)	Replace with forecast $\hat{y}_{T+h-i T}$

y_{T+h-i}	$h - 1 \leq 0$ (past value)	Use observed value y_{T+h-i}
ε_{T+h-i}	$h - 1 > 0$ (future value)	Replace with zero
ε_{T+h-i}	$h - 1 \leq 0$ (past value)	Replace with residual $\hat{\varepsilon}_{T+h-i}$

5.0 ANALYSIS AND RESULT

5.1. Univariate Modelling Technique

Three univariate exponential smoothing techniques were used on the daily closing price series for Google: Single Exponential Smoothing (SES), Holt's Method (branching into Double Exponential Smoothing), and the Adaptive response rate Exponential Smoothing (ARRES). The data includes daily closing prices for Google (Google / Alphabet Inc.) from January 2, 2013, to January 10, 2024 (2,775 total observations), and covered a price range from USD 17.51 to USD 150.71. An 11 year evaluation of the data in Figure 5.1 shows a pronounced upward trend in Google prices. The closing price series was transformed into a time series data structure of a periodic series of 252 trading days of the year.

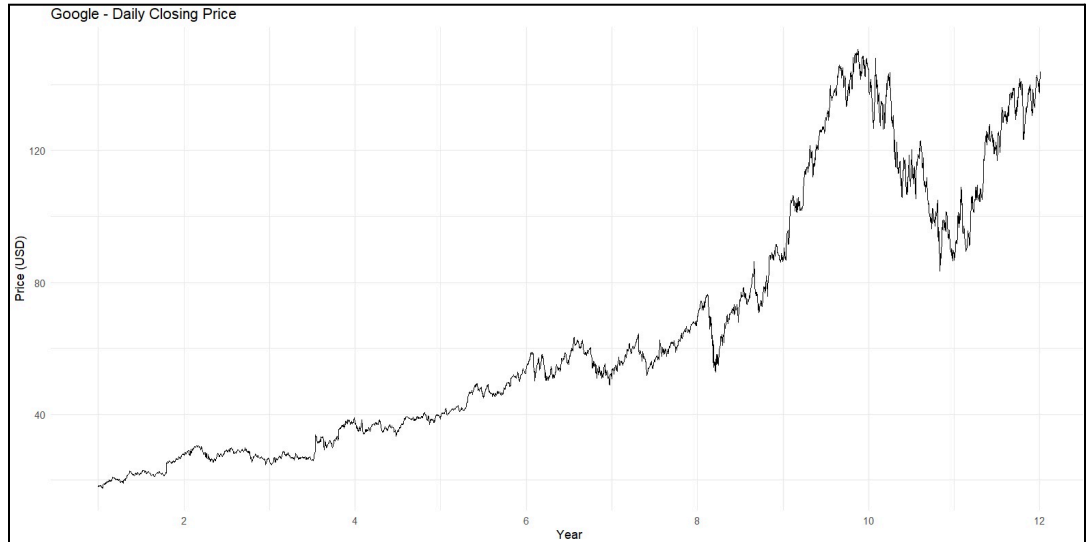


Figure 5.1: Google Daily Closing Price (2013 - 2024)

The dataset was divided into two parts, estimation and validation, for model building and testing. The estimation part, 70% of the total dataset ($n = 1,942$), was used to build the model and the remaining 30% ($n = 833$) was used for validation. The three models' parameters were also tuned based on the best fit values achieved by the estimation part of the dataset. The validation part of the dataset was actually used to test the models' forecasting abilities in the out-of-sample context. In order to maintain consistency/ fairness in the method of evaluation, out-of-sample forecasting abilities of all three models were evaluated based on one-step-ahead forecasts. Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) were the two metrics used to measure forecasting error and are defined as:

$$MSE = (1/n) \cdot \sum (actual - fitted)^2$$

$$MAPE = (1/n) \cdot \sum | (actual - fitted) / actual | \times 100\%$$

Model 1: Single Exponential Smoothing (SES)

SES captures only the level of a time series. SES performs well on time series data exhibiting neither seasonality nor trend, and can be viewed as the baseline benchmark. In R, to utilize SES, one can simply call the `ses()` function. If the user does not input a value for α , the SES function will attempt to find the best value for α that results in the minimal sum of squared errors (SSE) for the estimation part. The value of the first observation was used to set the first value of the level, l_0 .

Output 1: R Console Summary – SES Model

Table 5.1: Model Information for SES Model

Forecast method:	Simple exponential smoothing
Call:	ses(y = est_part, h = length(eva_part))
Smoothing parameters:	
alpha , α	0.8799
Initial states: l	18.0225
sigma:	0.7804

Table 5.2: Information Criteria for SES Model

AIC	AICc	BIC
13744.65	13744.66	13761.36

Table 5.3: Error Measures (Training Set) for SES Model

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.0333	0.7800	0.4763	0.0691	1.0744	0.0628	-0.0038

Table 5.4: Forecasts for SES Model

Point	Forecast	Lo 80	Hi 80	Lo 95	Hi 95
8.706349	74.94175	73.94166	75.94184	73.41224	76.47126
8.710317	74.94175	73.60961	76.27388	72.90443	76.97907
8.714286	74.94175	73.34520	76.53830	72.50003	77.38346
8.718254	74.94175	73.11874	76.76476	72.15369	77.72980

According to Output 1, $\alpha = 0.8799$ will be used as the optimal value for the smoothing constant, and $l_0 = 18.0225$ will be the initial value. Because stock prices exhibit an almost random walk, it appears that this model used the most recent observation almost exclusively, as indicated by the smoothing constant very close to 1.

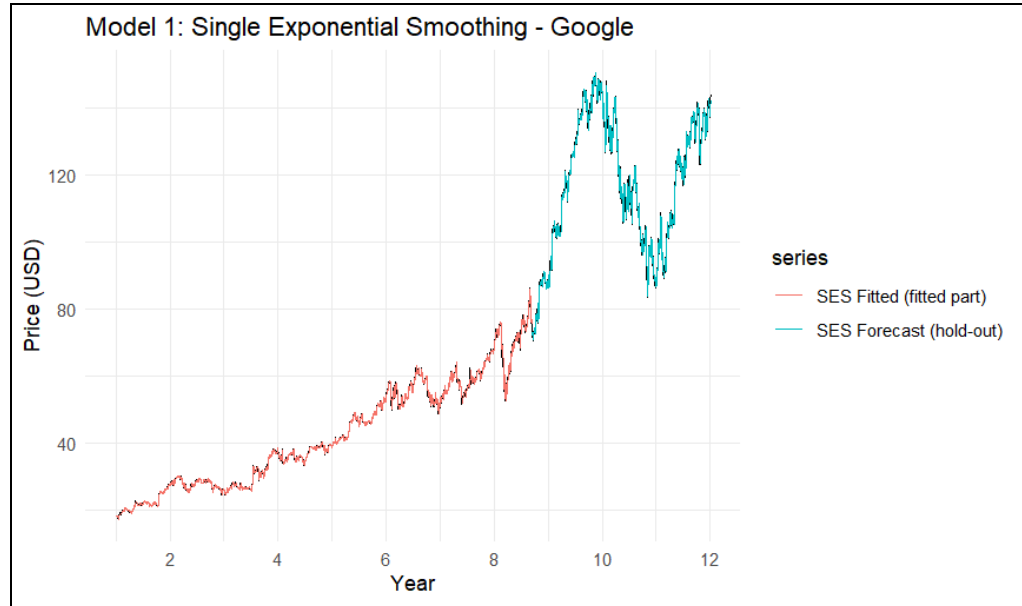


Figure 5.2: Model 1 SES Fitted vs Actual (Google)

Figure 5.2 shows Google's stock prices against the SES fitted values. The fitted values are closely related to the actual stock prices but are delayed by one day at the turning points in the data. Values of MSE and MAPE as shown in Appendix F for the estimation phase were 0.6084 and 1.0744% respectively. In the holdout phase, one-step ahead MSE and MAPE were 5.1038 and 1.4536%, respectively. The forecast value is USD 143.62

Proving the estimate values and forecast for the date 11/1/2024 using Single Exponential Smoothing Use $\alpha = 0.8799$

Date	Closing Price	Ft	E2t
03/01/2024	140.36	139.73	0.40
04/01/2024	138.04	140.28	5.02
05/01/2024	137.39	138.31	0.85
08/01/2024	140.53	137.50	9.18
09/01/2024	142.56	140.17	5.71
10/01/2024	143.80	142.27	2.34
11/01/2024		143.62	

Figure 5.3: Compute forecast value based on Excel

The Excel spreadsheet independently checks the Simple Exponential Smoothing (SES) forecast produced by the R script as shown in the Figure 5.3. A smoothing parameter of $\alpha = 0.8799$ is used to calculate the fitted value F_t and the squared error E_t^2 for each observation in the spreadsheet. The Excel calculation has a one-period-ahead forecast of USD143.62 for 11/01/2024, based on the historical closing prices up to 10/01/2024. This value is the same as the forecast obtained from the R script. Hence, the Excel spreadsheet is used to add supporting evidence to the correctness and consistency of the application of the SES forecasting procedure in the R script.

Model 2: Holt's Method (Double Exponential Smoothing)

SES is extended by Holt's Method with the inclusion of a trend component (β). Holt's Method is especially valuable for Google, given the existing data shows an eleven year upward trend of Google stock from USD 18 to USD 143. Both the initial level l_0 and initial trend b_0 are estimated by minimizing SSE with numerical optimization. The optimization of the smoothing constants α and β , is carried out by calling `holt()` in R, without the specification of the two parameters. In this case, Holt's Method is implemented, and R is allowed to determine the smoothing constants that minimize the sum of squared errors in the estimation of the dataset.

The values of α , β , l_0 , and b_0 are 0.8787, 0.0001, 18.5941, and 0.0288 respectively as shown in output 2. Since β is close to 0, the prediction for the next step will rely mostly on the level component of the forecast for this time series. Therefore for this time series, Holt's Method is basically equivalent to the Simple Exponential Smoothing (SES).

Output 2: R Console Summary – Holt's Method (Double Exponential Smoothing) Model

Table 5.5: Model Information for Holt's Method (Double Exponential Smoothing) Model

Forecast method:	Holt's method
Call:	<code>holt(y = est_part, h = length(eva_part))</code>

Smoothing parameters:	
alpha	0.8787
beta	1e-04
Initial states: l	18.5941
b	0.0288
sigma:	0.7802

Table 5.6: Information Criteria for Holt's Method (Double Exponential Smoothing) Model

AIC	AICc	BIC
13745.93	13745.96	13773.79

Table 5.7: Error Measures (Training Set) for Holt's Method (Double Exponential Smoothing) Model

	ME	RMSE	MAE	MPE	MAPE (%)	MASE	ACF1
Training set	0.0007	0.7794	0.4755	-0.0206	1.0755	0.0627	-0.0026

Table 5.8: Forecasts for Holt's Method (Double Exponential Smoothing) Model

Point	Forecast	Lo 80	Hi 80	Lo 95	Hi 95
8.706349	74.97644	73.97653	75.97635	73.44721	76.50567
8.710317	75.00540	73.67422	76.33658	72.96953	77.04127
8.714286	75.03436	73.43922	76.62949	72.59481	77.47391
8.718254	75.06332	73.24204	76.88459	72.27792	77.84872

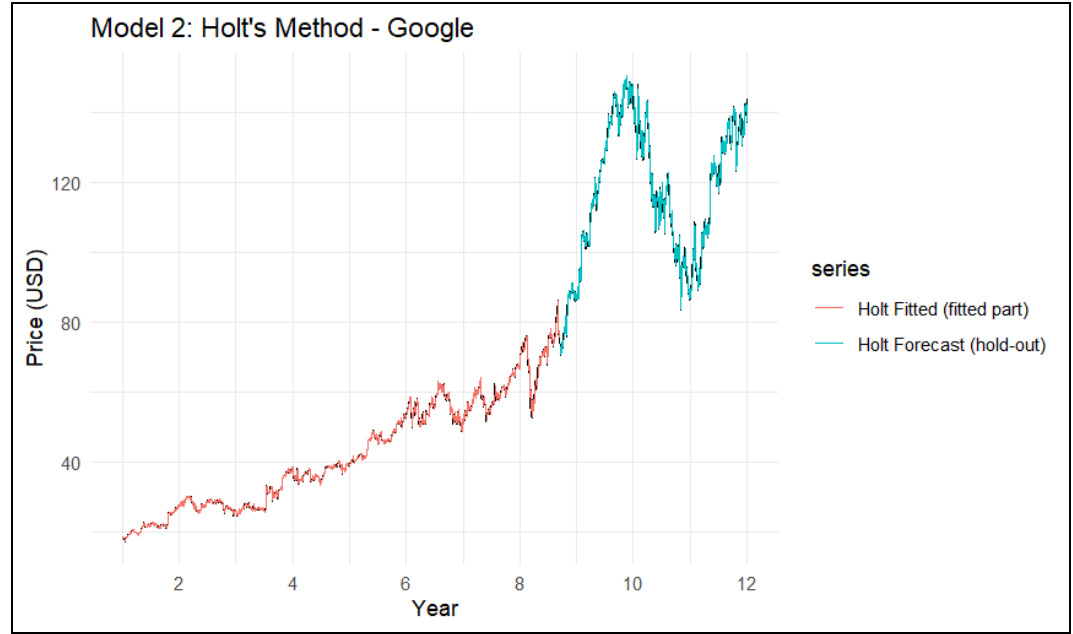


Figure 5.4: Model 2 Holt's Method Fitted vs Actual (Google)

Figure 5.4 shows the application of Holt's Method. Compared to the results of the SES Method, Holt's Method shows an even closer approximation to the actual values. Holt's Method is designed to account for the effects of the trend and level components in the fitted values, while the SES Method captures the level component only. In the estimation part, the obtained fitted values of Holt's Method as shown in Appendix G, reflected a MSE and MAPE of 0.6075 and 1.0755%, respectively. In the hold-out part, the one-step-ahead forecast of Holt's Method reflected a MSE and MAPE of 5.0963 and 1.4497%, respectively. The value forecast is USD 143.67.

Proving the estimate values and forecast for the date 11/1/2024 using Holt's Method (Double Exponential Smoothing) Use $\alpha = 0.88$, $\beta = 0.03$

Date	Closing Price	Lt	Tt	Ft	E2t
03/01/2024	140.36	140.30	0.14	139.89	0.22
04/01/2024	138.04	138.33	0.08	140.44	5.76
05/01/2024	137.39	137.51	0.05	138.41	1.04
08/01/2024	140.53	140.17	0.13	137.56	8.82
09/01/2024	142.56	142.29	0.19	140.30	5.11
10/01/2024	143.80	143.64	0.22	142.48	1.74
11/01/2024				143.86	

Figure 5.5: Compute forecast value based on Excel for Holt's Method (Double Exponential Smoothing)

The result of forecasting using the Holt's Method model is displayed in the Excel spreadsheet independently as shown in Figure 5.5. To calculate the level component L_t of the spreadsheet use the formula: $L_t = \text{average of } t \text{ observations}$, trend component T_t , fitted value F_t , and squared error E_t^2 for each observation. The Excel calculation results in a forecast for USD143.86 on 11/01/2024, based on historical prices up to 10/01/2024. This value is close to the value of USD143.67 that was obtained from the R script, the difference being only USD0.19. Note that the small deviation could be due to rounding at each intermediate step in the Excel calculation and differences in the number of digits retained in calculations with R. So, the Excel calculation gives supporting evidence that the Holt's Method forecasting procedure used in the R script is consistent and reasonable.

Model 3: Adaptive Response Rate Exponential Smoothing (ARRES)

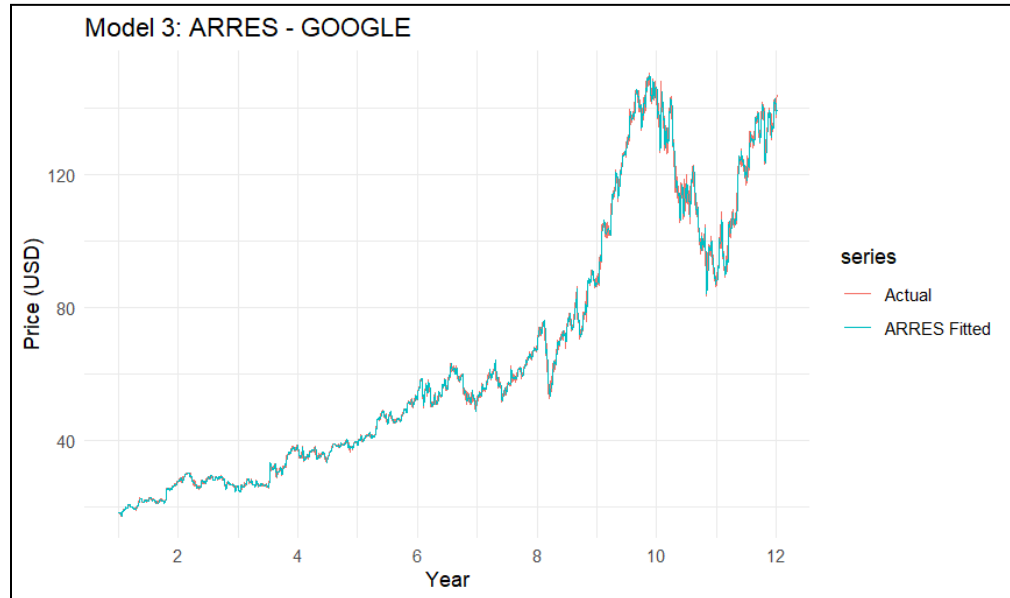


Figure 5.6: Model 3 ARRES Fitted vs Actual (Google)

As seen in Figure 5.6, ARRES is a self-adjusting model, in which, at each time point, α modifies itself according to the latest prediction errors. This is beneficial to the model in dealing with extreme variations in stock prices; for instance, sudden announcements of financial earnings or sudden declines in the stock market. Since α adjusts itself at each time point, there is no ‘best’ value for the parameter estimate for ARRES. There is no function for ARRES in R, the recursion explained in Chapter Two was coded manually:

$$\begin{aligned}
 e(t) &= y(t) - F(t) \\
 E(t) &= \beta \cdot e(t) + (1 - \beta) \cdot E(t-1) \\
 AE(t) &= \beta \cdot |e(t)| + (1 - \beta) \cdot AE(t-1) \\
 \alpha(t) &= |E(t) / AE(t)| \\
 F(t+1) &= \alpha(t) \cdot y(t) + (1 - \alpha(t)) \cdot F(t)
 \end{aligned}$$

Initial values: $F(1)=y(1)=18.0137$ (USD), $\alpha(1)=0.6$, $E(1)=AE(1)=0$, and $\beta=0.2$.

The value for the adaptive α at the conclusion of the estimation phase was 0.6533. As shown in Appendix H, The estimated part of the model ARRES for the mean squared error and the mean absolute percentage error were 0.8237 and 1.3336%, respectively. MSE and MAPE for the one-step ahead forecasting in the hold out sample were 7.4382 and 1.7953%, respectively.

Proving the estimate values and forecast for the date 11/1/2024 using Arres Use $\alpha = 0.6$, $\beta = 0.20$

Date	Price	Ft	et	Et	AEt	alpha	E2t
03/01/2024	140.36	141.35	-0.99	-0.34	1.39	0.24	0.98
04/01/2024	138.04	141.11	-3.07	-0.89	1.73	0.51	9.42
05/01/2024	137.39	139.54	-2.15	-1.14	1.81	0.63	4.62
08/01/2024	140.53	138.19	2.34	-0.44	1.92	0.23	5.48
09/01/2024	142.56	138.73	3.83	0.41	2.30	0.18	14.67
10/01/2024	143.80	139.42	4.38	1.20	2.72	0.44	19.18
11/01/2024		141.35					

Figure 5.7: Compute forecast value based on Excel for ARRES

The results of the ARRES forecasting model are checked independently in the Excel spreadsheet as shown in Figure 5.7. The fitted values F_t , the forecast errors e_t , the

smoothed error E_t , the smoothed absolute error AE_t , the adaptive smoothing parameter α , and the squared error E_t^2 for each observation are calculated in the spreadsheet. The Excel calculation gives a forecast of USD141.35 for 11/01/2024, based on the historical data up to 10/01/2024. This is very close to the forecast produced by the R script which was USD141.37. This is due to the rounding of each intermediate value in the excel calculation to two decimal places as opposed to greater numerical precision in R. Thus the result of the Excel model is supportive evidence of the correct and reliable forecasting process used in the R script, ARRES.

Model Comparison

Table 1 reports the MSE and MAPE for all three models on both the estimation and hold-out parts.

Table 5.9: Error measures for the three univariate models

Model	Parameters	Fitted MSE	Fitted MAPE (%)	Hold-out MSE	Hold-out MAPE (%)
SES	$\alpha = 0.8799$	0.6084	1.0744	5.1038	1.4536
Holt's Method	$\alpha = 0.8787$, $\beta = 0.0001$	0.6075	1.0755	5.0963	1.4497
ARRES	Adaptive α ($\beta = 0.2, \alpha_1 = 0.6$)	0.8237	1.3336	7.4382	1.7953

Table 5.9 reveals the MSE and MAPE error values for each method in the estimation phase. Errors in the SES method are indicated as MSE = 0.6084 and MAPE = 1.0744. For the Holt method, errors are indicated as MSE = 0.6075 and MAPE = 1.0755. The ARRES method has the highest values, with MSE = 0.8237 and MAPE = 1.3336. These results show that the ARRES method, with its adaptive smoothing technique, is the most flexible method in the estimation

phase, while the other methods, with their fixed parameters, are more rigid in comparison.

For the hold-out phase, the results show that the three methods have increasingly greater differences. Errors for the Holt method show $MSE = 5.0963$ and $MAPE = 1.4497$, while for the SES method $MSE = 5.1038$ and $MAPE = 1.4536$. In this case, Holt's method is superior to the SES method, and the difference is so small it is only visible beyond the fourth decimal place. A likely cause for Holt's method behaving almost the same as SES in this case, which resembles a near random walk, is the likely estimation of the trend parameter $\beta = 0.0001$ (i.e., $\beta = 0$).

ARRES performs the worst in the hold-out period, with an MSE of 7.4382 and $MAPE$ of 1.7953%. ARRES performs about 46% and 24% worse, respectively, than Holt's Method. Although ARRES is built with the idea that the model will best deal with the 'unexpected' price movements because of the time-variant α , ARRES performs the worst with this data. The Google data, despite being jumpy, does not contain the 'unexpected' types of changes in structures that ARRES is designed to deal with. Thus, in the hold-out period, the SES and Holt's Method Models perform better.



Figure 5.8: Hold-out Period Actual vs All Three Models

According to Figure 5.8, within the hold-out timeframe, SES and Holt's Method both obviously outperform ARRES. While they perform comparably, Holt's Method edges SES with a marginal (yet insignificant) reduction in hold-out MSE to 5.0963 and hold-out MAPE to 1.4497%, with SES at 5.1038 and 1.4536%, respectively. Holt's Method also outperforms ARRES, which is the least of the three in effectiveness in her adaptive smoothing.

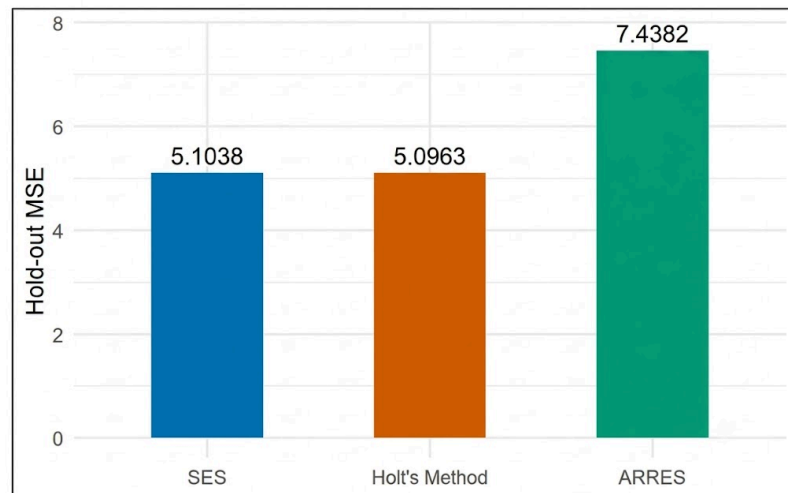


Figure 5.9: Hold-out MSE by Univariate Model

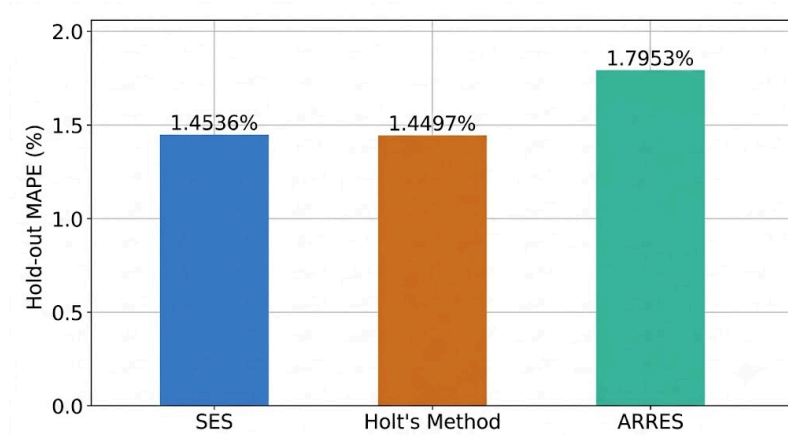


Figure 5.10: Hold-out MAPE by Univariate Model

Figures 5.9 and 5.10 illustrate the hold-out MSE and MAPE for each model. ARRES records a hold-out MSE of 7.4382, approximately 46% higher than Holt's Method (5.0963), and a hold-out MAPE of 1.7953% compared to 1.4497% for Holt's Method. The forecast value is USD 141.37.

Best Univariate Model and One-Step-Ahead Forecast

Based on the hold-out MSE and MAPE, Holt's Method is the best univariate model in this case. Among the three models, Holt's Method produced the best results, with a hold-out MSE of 5.0963 and a hold-out MAPE of 1.4497%. It should be noted that, with a trend parameter, β , of 0.0001, Holt's Method becomes very similar to SES. In this case, Holt's Method would not be the better model as the forecasts from both methods would be nearly identical. However, Holt's Method is the more appropriate model to choose as it slightly better minimized errors in the hold-out period for both MSE and MAPE.

Differing from the fixed-parameter models in this series, ARRES demonstrated no benefits from its propensity to change the smoothing constant during each time step. The Google daily closing price series acts primarily as a random walk. The most reliable predictor of tomorrow's price is today's price. In this instance, a model that employs a high fixed α would be a better predictor as compared to a model that would be variable due to a changing α as a result of error. In addition, ARRES performed the most poorly of the three models,

showing a substantially greater hold-out MSE (7.4382) and MAPE (1.7953%) when compared to SES and Holt's Method, for one-step-ahead forecasting of this series.

Based on Holt's Method as the best model and using the entire 2,775 observation data series, one-step-ahead forecasts predict the next day's trading closed at USD 143.67, compared to the most recent close of USD 143.80. The forecast close price of USD 143.67 reflects model confidence and proximity to the most recently observed price due to a high model α (0.8787). A high α value indicates that the model has placed considerable confidence on the most recent observation and may alter the forecast minimally. In this context, the same trading day forecasts by SES and ARRES, at USD 143.62 and USD 141.37, respectively, indicate that the SES and Holt's Method approaches are similar for this forecasting series. Holt's Method and SES may have a recent closing price bias, while ARRES and SES, with a α of 0.4456 and a forecast of USD 141.37, are closing the gap to a smoothed history of the closing prices.

5.2. ARIMA Models

PART A : MODEL IDENTIFICATION

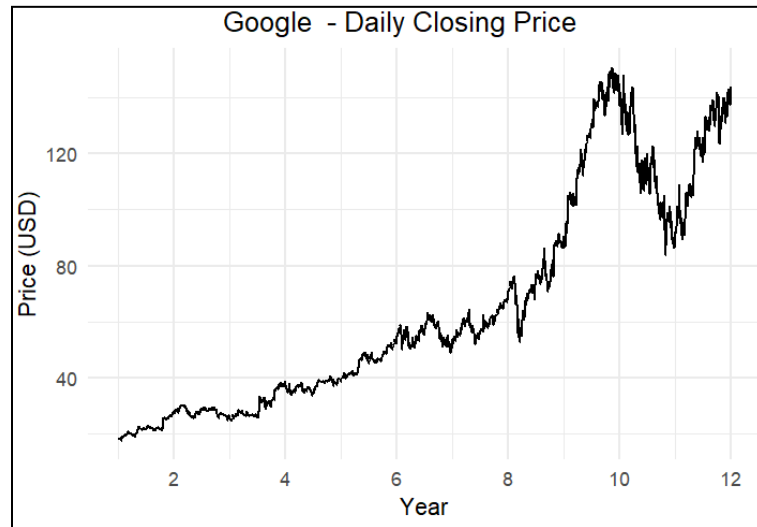


Figure 5.11 show the time series plot of Google Daily Closing Price

Based on Figure 5.11, the series of Google daily closing prices is non-stationary. The series has a constant upward trend throughout the entire observation period, suggesting the series likely has a non-constant mean. With a non-constant mean, we can suspect a non-constant variance. The series has a trend and does not satisfy the stationary condition of constant mean and variance.

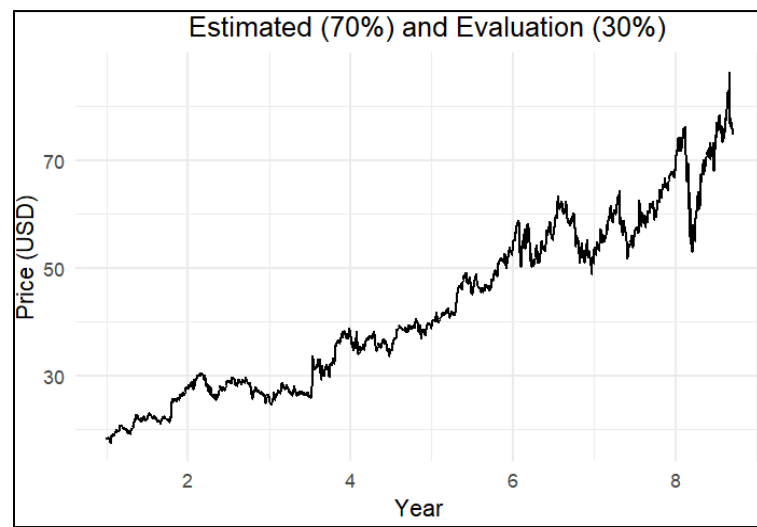


Figure 5.12: show the Google Daily Closing Price: Estimation (70%) and Evaluation (30%) Periods

The daily closing price data for Google was divided into 70% estimation data and 30% evaluation data in chronological order. Here, the first 70% of the data is examined and the pattern of the data series is learned, while the remaining 30% is reserved to test the forecasting technique on the data that comes after the estimation. Analyzing the behavior of the data typically requires a larger estimation portion. In contrast, the evaluation portion is used to assess forecast accuracy. Based on the trend of the data, an ARIMA model can be possibly applied.

Then the plot will be tested with 4 tests, which are the ADF test, PP test, KPSS test, and ADF plot, to support whether the data are stationary or not:

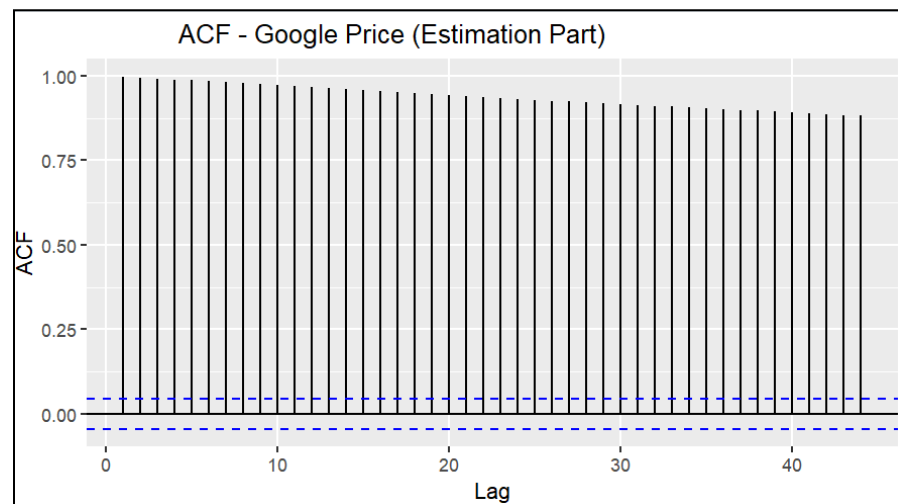


Figure 5.13 shows the ACF plot visualization.

Table 5.10: Stationarity Test for Google Closing Price

Method	Result	Interpretation
ACF plot	The autocorrelation coefficients remain close to 1 and decay very slowly across many lags.	A stationary series should have an ACF that decays rapidly to zero. Therefore, the slow decay indicates that the Google price series is non-stationary.
ADF test	Statistic = -3.817 ; p-value = 0.01813	Reject H_0 because p-value < 0.05 . The ADF test indicates that the series is stationary.
PP test	Statistic = -33.641 ; p-value < 0.01	Reject H_0 because p-value < 0.05 . The PP test indicates that the series is stationary.
KPSS test	Statistic = 20.605 ; p-value < 0.01	Reject H_0 because p-value < 0.05 . The KPSS test indicates that the series is non-stationary.

The results are conflicting because the ADF and PP tests indicate stationarity, while the KPSS test and ACF plot indicate non-stationarity. Since the ACF coefficients decay very slowly and the original series contains a clear trend, the series is treated as non-stationary. Therefore, first-order differencing is performed before achieving stationarity.

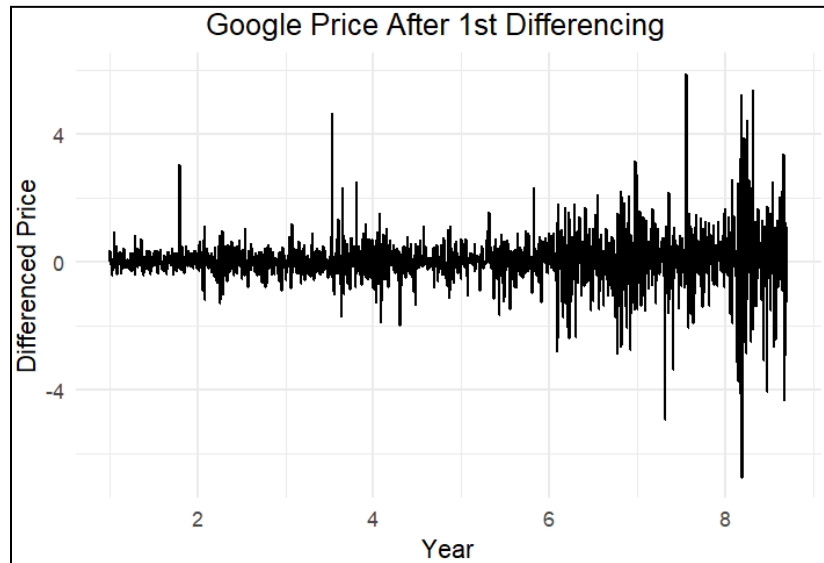


Figure 5.14 show the time series plot of Google's daily closing price after first-order differencing.

After initial differencing, the upward trend in the series is eliminated. The differenced values are shown to vary about zero. This indicates that the mean value of the series is now more consistent. However, after around Year 6, the differences display more variation and include large extreme positive and negative values. First differencing helps to meet the stationarity requirement. However, uncertainty remains about the effect of differencing on the stationarity of the series and the ACF and other tests for stationarity should be performed.

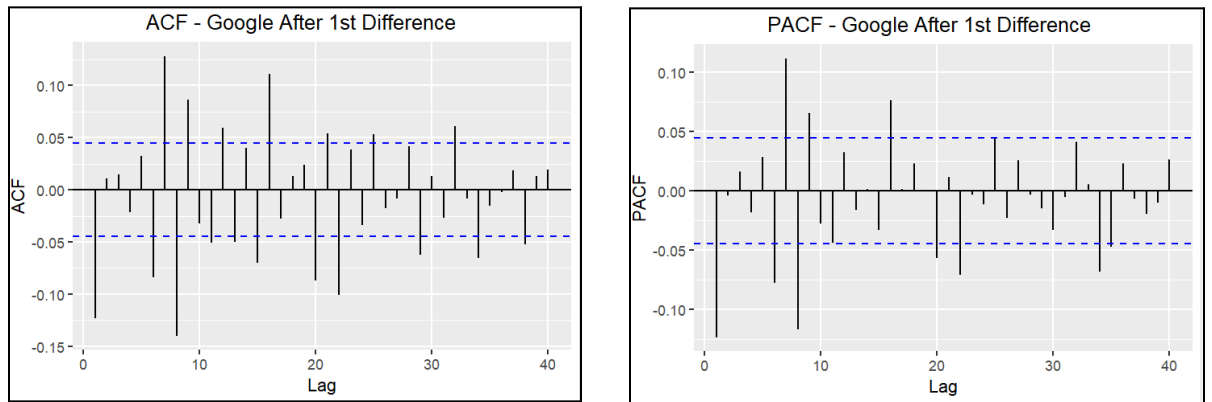


Figure 5.15 shows the ACF and PACF plots of the Google closing price after first-order differencing.

Proving the standard error of autocorrelation is where $T = 1941$ and the confidence limit are 95%;

$$SE(r_k) = \pm 1.96 \frac{1}{\sqrt{1941}} \\ \pm 0.0455$$

Therefore:

$$-0.0445 \leq r_k \leq 0.0445$$

Table 5.11: Stationarity Test for Google Closing Price after First-Order Differencing

Method	Result	Interpretation
ACF plot	The autocorrelation coefficients fluctuate around zero. Most spikes die off quickly to zero. There is no slow decay across the lags and also a significant spike at many lags.	The significant spike suggests that the differenced series is stationary.
PACF plot	The partial autocorrelation coefficients also fluctuate around zero, with several significant spikes but no persistent pattern.	The PACF supports the view that the strong dependence found in the original series has been reduced after differencing.
ADF test	Test statistic = -12.557 ; p-value < 0.01	Reject H_0 because p-value < 0.05. The differenced series is stationary.
PP test	Test statistic = -2169.3 ; p-value < 0.01	Reject H_0 because p-value < 0.05. The differenced series is stationary.
KPSS test	Test statistic = 0.032092; p-value = 0.10	Fail to reject H_0 because p-value > 0.05. The differenced series is stationary.

After first-order differencing, the slow-decaying patterns that occur in the original series are no longer present in the ACF and PACF. The coefficients now fluctuate about zero and indicate that the correlation that existed among distant points diminished. The ADF and PP tests indicate that a unit root is present, while the null hypothesis for the KPSS is not rejected indicating that the series is stationary. Thus, all three formal tests confirm that the Google price series is stationary after first-order differencing and that second-order differencing is not necessary.

Based on the ACF and PACF, five (5) models were determined to be appropriate for this data set. This is because in order to identify the best-fitted model, the d parameter is constrained to be equal to $d = 1$.

Table 5.12: The table shows how the parameter of ARIMA is achieved.

Model	Parameter	Explanation
Model 1	ARIMA(1,1,0)	PACF has a most significant spike at lag 1, suggesting an AR(1) component.
Model 2	ARIMA(0,1,1)	ACF has a most significant spike at lag 7, suggesting an MA(1) component.
Model 3	ARIMA(1,1,1)	Both ACF and PACF have significant spikes at lag 1.
Model 4	ARIMA(2,1,2)	Included as a broader low-order competing model because both plots contain several significant spikes without a clear cutoff.
Model 5	Auto ARIMA (3,1,2)	Used as an automatically selected benchmark model and compared with the four manually identified models.

PART B : MODEL ESTIMATION AND VALIDATION

MODEL ESTIMATION

The five candidate models were estimated using the 70% estimation and 30% evaluation of the Google closing-price data, producing an estimated backshift equation based on the coefficients value.

Table 5.13: Show the model equation backshift before and after coefficients

Model	Backshift Equation with Coefficients
ARIMA(1,1,0)	$(1 + \Phi_1 B)(1 - B)y_t = c + \varepsilon_t$ $(1 + 0.1234B)(1 - B)y_t = 0.0329 + \varepsilon_t$

ARIMA(0,1,1)	$(1 - B)y_t = c + (1 - \theta_1 B)\varepsilon_t$ $(1 - B)y_t = 0.0293 + (1 - 0.1221B)\varepsilon_t$
ARIMA(1,1,1)	$(1 + \Phi_1 B)(1 - B)y_t = c + (1 - \theta_1 B)\varepsilon_t$ $(1 + 0.0951B)(1 - B)y_t = 0.0321 + (1 - 0.0287B)\varepsilon_t$
ARIMA(2,1,2)	$(1 + \Phi_1 B + \Phi_2 B^2)(1 - B)y_t = c + (1 + \theta_1 B + \theta_2 B^2)\varepsilon_t$ $(1 + 0.2881B + 0.1168B^2)(1 - B)y_t =$ $0.0411 + (1 + 0.1639B + 0.0943B^2)\varepsilon_t$
Auto ARIMA (3,1,2)	$(1 + \Phi_1 B + \Phi_2 B^2 + \Phi_3 B^3)(1 - B)y_t = (1 + \theta_1 B + \theta_2 B^2)\varepsilon_t$ $(1 + 1.8184B + 1.0009B^2 + 0.0407B^3)(1 - B)y_t$ $= (1 + 1.7335B + 0.8576B^2)\varepsilon_t$

Table 5.14 : Show estimation results using the 70% data

Model	δ	AIC	AICc	BIC	RMSE	MAE	MAPE
ARIMA(1,1,0)	0.6081	4546.83	4546.85	4563.55	0.779	0.475	1.07
ARIMA(0,1,1)	0.6082	4547.12	4547.14	4563.84	0.779	0.475	1.07
ARIMA(1,1,1)	0.6084	4548.80	4548.83	4571.09	0.779	0.475	1.07
ARIMA(2,1,2)	0.6089	4552.37	4552.41	4585.79	0.779	0.475	1.07
Auto ARIMA(3,1,2)	0.5876	4483.72	4483.76	4517.14	0.765	0.478	1.08

Table 5.15: Show evaluation of results using the 30% data

Model	AIC	BIC	RMSE	MAE	MAPE	ACF1
ARIMA(1,1,0)	3720.35	3725.07	2.26	1.66	1.45	0.109
ARIMA(0,1,1)	3719.77	3724.49	2.26	1.66	1.45	0.108
ARIMA(1,1,1)	3720.34	3725.07	2.26	1.66	1.45	0.109
ARIMA(2,1,2)	3721.71	3726.44	2.26	1.67	1.45	0.109
Auto ARIMA(3,1,2)	3737.49	3742.21	2.28	1.68	1.47	0.0463

MODEL VALIDATION

Model validation was conducted using the residuals obtained from the five models estimated with the 70% estimation data. The Ljung–Box test was applied at 44 lags to determine whether the residuals were white noise.

Prove lag 44: if $n = 1941$, the lag is $\sqrt{1941}$

Table 5.16: Ljung–Box Test Results for the Estimated Models

Model	Ljung–Box statistic	df	p-value	Decision	Conclusion
ARIMA(1,1,0)	189.16	44	$P < 2.2 \times 10^{-16}$	Reject H_0	Residuals are not white noise
ARIMA(0,1,1)	194.40	44	$P < 2.2 \times 10^{-16}$	Reject H_0	Residuals are not white noise
ARIMA(1,1,1)	189.98	44	$P < 2.2 \times 10^{-16}$	Reject H_0	Residuals are not white noise
ARIMA(2,1,2)	190.54	44	$P < 2.2 \times 10^{-16}$	Reject H_0	Residuals are not white noise

Auto ARIMA(3,1,2)	79.136	44	0.0009093	Reject H_0	Residuals are not white noise
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All five models produced p-values below 0.05. Therefore, the null hypothesis was rejected for every model, indicating that their residuals still contain significant autocorrelation and are not white noise. Consequently, none of the five models fully satisfies the model-validation requirement.

Although Auto ARIMA(3,1,2) has the smallest Ljung–Box statistic among the models, its p-value of 0.0009093 remains below 0.05. Therefore, it also cannot be considered a well-specified model. Based on the Box–Jenkins procedure, additional candidate models should be identified and estimated before a final model is selected. Then choose the best model using AIC, AICc, and BIC criteria.

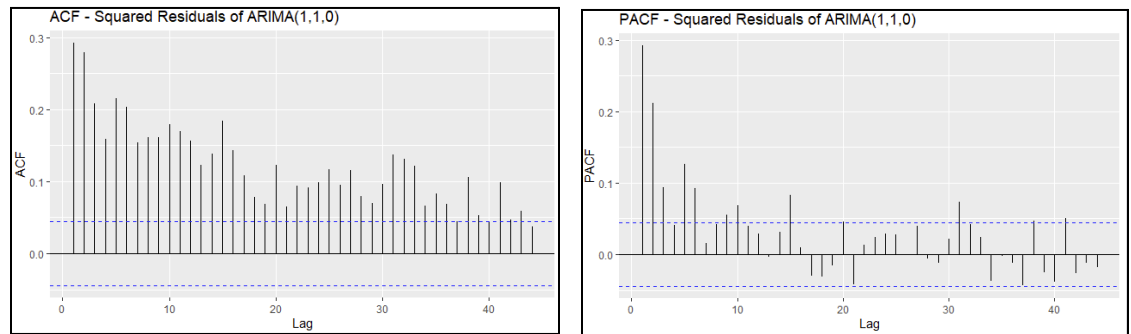


Figure 5.16 shows the ACF and PACF plots of the squared residual of ARIMA (1,1,0)

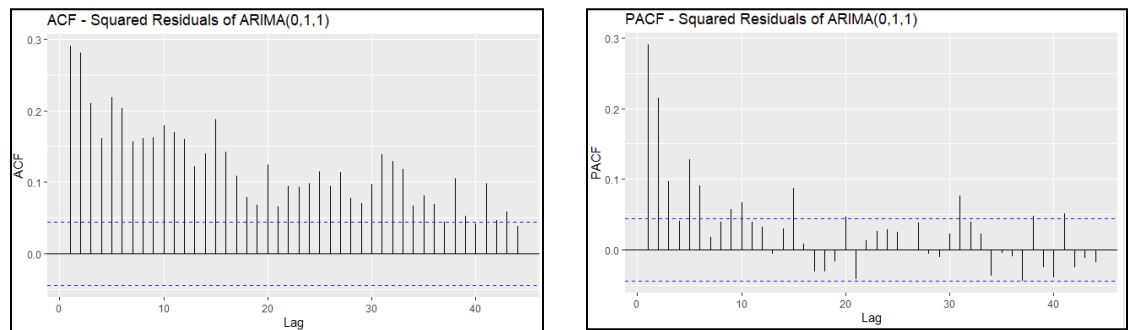


Figure 5.17 shows the ACF and PACF plots of the squared residual of ARIMA (0,1,1)

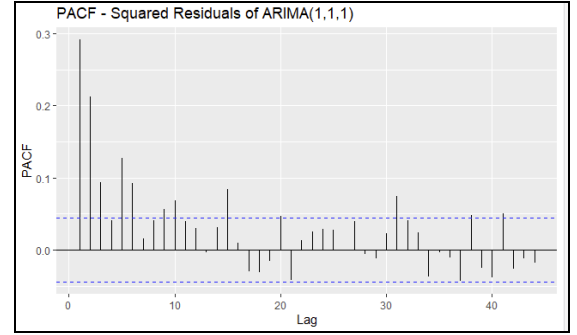
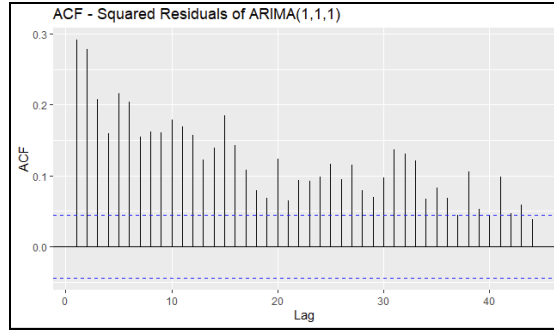


Figure 5.18 shows the ACF and PACF plots of the squared residual of ARIMA (1,1,1)

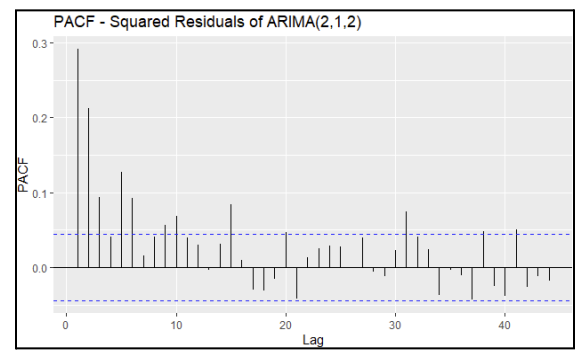
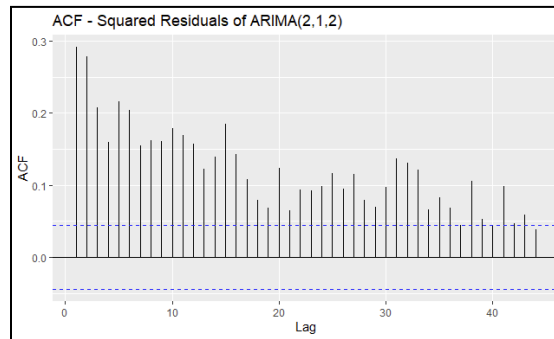


Figure 5.19 shows the ACF and PACF plots of the squared residual of ARIMA (2,1,2)

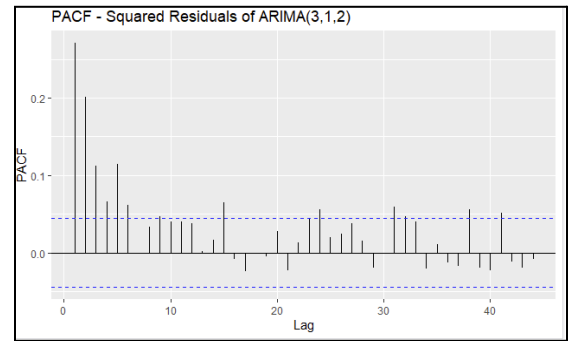
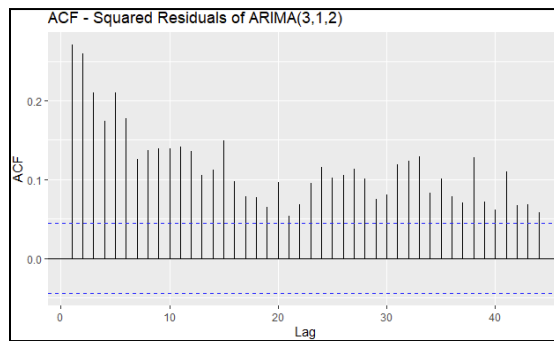


Figure 5.20 shows the ACF and PACF plots of the squared residual of ARIMA (3,1,2)

Based on the observations of figures 5.16, 5.17, 5.18, 5.19 and 5.20 that can be included, there is a significant spike observed in both ACF and PACF. Hence, the residuals are not white noise.

Better model fit is indicated by lower values of AIC, AICc, and BIC. AICc is a sample size adjustment of AIC, and BIC is a more substantial penalty for complexity adjustment of AIC.

Table 5.17: Comparison of AIC, AICc and BIC Values for the Five Candidate ARIMA Models

Rank	Model	AIC	AICc	BIC
1	Auto ARIMA(3,1,2)	4483.72	4483.76	4517.14
2	ARIMA(1,1,0)	4546.83	4546.85	4563.55
3	ARIMA(0,1,1)	4547.12	4547.14	4563.84
4	ARIMA(1,1,1)	4548.80	4548.83	4571.09
5	ARIMA(2,1,2)	4552.37	4552.41	4585.79

According to AIC and AICc, ARIMA (3, 1, 2) is the best model because the value is the lowest between the 5 models. However, based on the BIC criterion, it also shows the BIC of ARIMA (3,1,2) is also the lowest, meaning that is the best model. Auto ARIMA(3,1,2) recorded the value of AIC of 4483.72, AICc of 4483.76, and BIC of 4517.14. Hence, it was selected as the best-fitting model among the five candidates.

MODEL APPLICATION

The next one-step-ahead forecast based on ARIMA has been chosen, which is ARIMA (3,1,2).

Backshift Equation from table 5.13:

$$\begin{aligned}
 (1 + 0.8184B - 0.8175B^2 - 0.9602B^3 - 0.0407B^4)y_t \\
 = (1 + 1.7335B + 0.8576B^2)\varepsilon_t
 \end{aligned}$$

Using:

$$B^k y_t = y_{t-k}$$

Change t and place y_t alone on the left:

$$y_t = -0.8184 y_{t-1} + 0.8175 y_{t-2} + 0.9602 y_{t-3} + 0.0407 y_{t-4} \\ + \varepsilon_t + 1.7335 \varepsilon_{t-1} + 0.8576 \varepsilon_{t-2}$$

Replace t with $T + 1$:

$$y_{T+1} = -0.8184 y_T + 0.8175 y_{T-1} + 0.9602 y_{T-2} + 0.0407 y_{T-3} \\ + \varepsilon_{T+1} + 1.7335 \varepsilon_T + 0.8576 \varepsilon_{T-1}$$

Since ε_{T+1} is a future error:

$$E(\varepsilon_{T+1}) = 0$$

Thus, the one-step-ahead forecast is:

$$\hat{y}_{T+1|T} = -0.8184 y_T + 0.8175 y_{T-1} + 0.9602 y_{T-2} + 0.0407 y_{T-3} \\ + 1.7335 e_T + 0.8576 e_{T-1}$$

The price of the closing price of yearly data is based on the dataset of Google's 10-year stock prices. A dataset for 2772 observations produced an output of ARIMA (3, 1, 2) as follows. Because 6 and 7 January 2024 were non-trading days, the previous available observation before 8 January was 5 January 2024.

For the forecast of 11 January 2024:

Therefore:

- $Y_t = 10 \text{ January } 2024$
- $Y_{t-1} = 9 \text{ January } 2024$

- Y_{t-2} = 8 January 2024
- Y_{t-3} = 5 January 2024

Table 5.18: Recent Google Closing Prices, Fitted Values and Residuals

Time	Closing Price	Fitted	Residual
5/1/2024	137.39	138.35	-0.96
8/1/2024	140.53	138	2.53
9/1/2024	142.56	140.06	2.50
10/1/2024	143.80	142.45	1.35

Proving that using an equation that shows the closing price of 11/1/2024 is 143.85 based on the forecast has been achieved through R.

```

Series: value
Model: ARIMA(3,1,2)

Coefficients:
          ar1      ar2      ar3      ma1      ma2
      -1.8184  -1.0009  -0.0407   1.7335   0.8576
s.e.    0.0492   0.0652   0.0307   0.0430   0.0322

sigma^2 estimated as 1.977:  log likelihood=-4881.45
AIC=9764.9   AICC=9764.9   BIC=9770.83
>

```

Figure 5.21 shows the report for ARIMA (3,1,2)

```

> print(forecast_table, row.names = FALSE)
Point Forecast
 1    143.85
 2    143.72
 3    143.86
 4    143.73
 5    143.82
 6    143.77
 7    143.78
 8    143.82
 9    143.74
10    143.84

```

Figure 5.22 shows the forecast value of 10 days ahead for ARIMA (3,1,2)

Therefore, substitute all the value:

$$\begin{aligned}\hat{y}_{11/1/2024} = & - 0.8184(143.80) + 0.8175(142.56) \\ & + 0.9602(140.53) + 0.0407(137.39) \\ & + 1.7335(1.35) + 0.8576(2.50)\end{aligned}$$

Calculate each term:

$$\begin{aligned}\hat{y}_{11/1/2024} = & - 117.6859 + 116.5428 \\ & + 134.9369 + 5.5918 \\ & + 2.3402 + 2.1440\end{aligned}$$

Therefore:

$$\hat{y}_{11/1/2024} = 143.87$$

The ARIMA(3,1,2) model was applied to produce a one-step-ahead forecast for the Google closing price . Using the rounded coefficients, closing prices and residuals, the manual forecast was approximately USD 143.87. The R output produced a more precise forecast of USD 143.85 in R , which was rounded to USD 143.85. The slight difference was caused by rounding, as the software used the full coefficient and residual values. Basically this proves why the value of the forecast USD 143.86

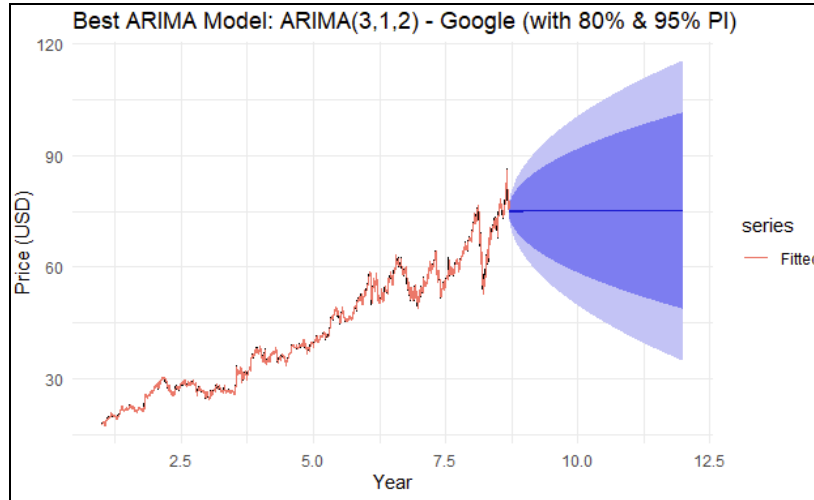


Figure 5.23 shows the plot forecast for ARIMA (3,1,2)

The ARIMA(3,1,2) model produces both historical fitted values and future predictions for Google's daily closing prices in Figure 5.23. The red fitted line captures the historical price trend. The model predicts the price will remain stable and close to the last predicted value at USD 75, in the approximate horizontal line at the beginning of the forecast period. The region with shading represents an 80% prediction interval and the lighter region represents a 95% prediction interval. The prediction intervals are wider at the long forecast horizons, showing higher uncertainty for longer-term predictions. The central forecasts, in contrast, remain stable. The model predicts price points, but wide prediction intervals show that the actual future prices may differ significantly from the predicted values.

Comparison Between the Best Univariate and ARIMA Models

In order to select the most suitable forecasting model for the Google daily closing price series, the leading candidate from the univariate models was compared to the leading candidate from the ARIMA models. The comparison utilized two forecast error metrics: Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE). A lower value reflects superior forecasting efficiency.

Table 5.19: Hold-Out Error Comparison Between Holt's Method and ARIMA(3,1,2)

Error measure	Holt's Method	Auto ARIMA(3,1,2)	Better model
MSE	5.0963	5.1984	Holt's Method
MAPE	1.4497%	1.4700%	Holt's Method

The MSE for ARIMA(3,1,2) was calculated from its reported RMSE:

$$MSE = (RMSE)^2$$

$$MSE = (2.28)^2 = 5.1984$$

Since the RMSE value was rounded to two decimal places, the ARIMA MSE of 5.1984 is an approximate value.

The best univariate forecasting model, Holt's Method, and the best ARIMA model, Auto ARIMA(3,1,2), were compared using the MSE and MAPE results from the 30% holdout sample. Holt's Method produced an MSE of 5.0963 and MAPE of 1.4497%. In comparison, Auto ARIMA(3,1,2) produced an approximate MSE of 5.1984 and MAPE of 1.4700%. With Holt's Method producing the best results in both cases, it is noted that forecasting with Holt's Method will be superior. Holt's Method is also much simpler and more parsimonious than ARIMA(3,1,2). The residuals of the ARIMA model also failed the Ljung-Box test, confirming that the residuals are not white noise. For all the reasons stated, Holt's Method is the best forecasting method for the Google daily closing price series.

6.0 CONCLUSION

This study successfully achieved all three research objectives in analyzing and forecasting the daily closing prices of Google. For the first objective, the analysis identified a clear long-term upward trend, medium-term fluctuations. No consistent seasonal pattern was observed, while the strong influence of the latest observation suggested that the stock price behaved approximately like a random-walk process.

For the second objective, several forecasting approaches were evaluated, including Single Exponential Smoothing, Holt's Method, Adaptive Response Rate Exponential Smoothing and five ARIMA candidates. Holt's Method was identified as the best univariate model, recording the lowest hold-out MSE of 5.0963 and MAPE of 1.4497%. Within the ARIMA category, Auto ARIMA(3,1,2) produced the lowest AIC, AICc and BIC values. However, it failed the Ljung–Box residual test, indicating that some autocorrelation remained unexplained. It also produced slightly higher hold-out errors, with an MSE of 5.1984 and a MAPE of 1.4700%.

For the third objective, Holt's Method was applied to the complete dataset to generate a one-step-ahead forecast of USD 143.67. Its forecasting accuracy was evaluated using the 30% hold-out sample, providing evidence that the model performed consistently on observations not used for initial estimation. The high smoothing parameter, $\alpha=0.8787$, further shows that the model placed substantial weight on the most recent closing price, which was appropriate for the observed behavior of the series.

Overall, Holt's Method was selected as the final forecasting model because it provided the best combination of accuracy, simplicity, and parsimony. Therefore, the study fulfilled its intended objectives by identifying the behavior of the Google closing price series, selecting the most suitable forecasting model, and producing an evidence-based forecast for the next trading period. Nevertheless, the forecast represents an expected value based on historical data and should not be interpreted as a guaranteed future price.

7.0 REFERENCE

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8.0 APPENDICES

```
> forecast(ses(googlets, alpha = alpha_ses), h = 1)
      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
12.0119      143.6166 141.822 145.4111 140.8721 146.3611
```

Appendix A: One-Step Ahead For SES

```
> forecast(holt(googlets, alpha = alpha_holt, beta = beta_holt), h = 1)
      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
12.0119      143.6661 141.8718 145.4604 140.922 146.4102
```

Appendix B: One-Step Ahead For Holt's Method

```
> last_alpha <- tail(arres_full$alpha, 1)
> last_fitted <- tail(arres_full$fitted, 1)
> last_actual <- tail(as.numeric(googlets), 1)
> arres_onestep <- last_alpha * last_actual + (1 - last_alpha) * last_fitted
> last_alpha * last_actual + (1 - last_alpha) * last_fitted
[1] 141.3699
```

Appendix C: One-Step Ahead For Arres

```
Forecast method: Simple exponential smoothing

Model Information:
Simple exponential smoothing

Call:
ses(y = est_part, h = length(eva_part))

Smoothing parameters:
alpha = 0.8799

Initial states:
l = 18.0225

sigma: 0.7804

      AIC      AICc      BIC
13744.65 13744.66 13761.36

Error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.03330957 0.7799743 0.476299 0.06906751 1.074368 0.06283158 -0.003800194

Forecasts:
      Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
8.706349      74.94175 73.94166 75.94184 73.41224 76.47126
8.710317      74.94175 73.60961 76.27388 72.90443 76.97907
8.714286      74.94175 73.34520 76.53830 72.50003 77.38346
8.718254      74.94175 73.11874 76.76476 72.15369 77.72980
```

Appendix D: Simple Exponential Smoothing Model Summary and Forecasts

```

Forecast method: Holt's method

Model Information:
Holt's method

Call:
holt(y = est_part, h = length(eva_part))

Smoothing parameters:
  alpha = 0.8787
  beta  = 1e-04

Initial states:
  l = 18.5941
  b = 0.0288

sigma: 0.7802

      AIC      AICc      BIC
13745.93 13745.96 13773.79

Error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.0006870451 0.7794284 0.4754636 -0.02056929 1.075547 0.06272137 -0.002577152

Forecasts:
      Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
8.706349      74.97644 73.97653 75.97635 73.44721 76.50567
8.710317      75.00540 73.67422 76.33658 72.96953 77.04127
8.714286      75.03436 73.43922 76.62949 72.59481 77.47391
8.718254      75.06332 73.24204 76.88459 72.27792 77.84872

```

Appendix E: Holt's Linear Trend Model Summary and Forecasts

```

> calc_MSE(est_part, ses_fitted)
[1] 0.6083599
> calc_MAPE(est_part, ses_fitted)
[1] 1.074368
> calc_MSE(eva_part, ses_forecast)
[1] 5.103815
> calc_MAPE(eva_part, ses_forecast)
[1] 1.453603
>

```

Appendix F: Forecast Evaluation and Error Metrics for Simple Exponential Smoothing

```

> calc_MSE(est_part, holt_fitted)
[1] 0.6075086
> calc_MAPE(est_part, holt_fitted)
[1] 1.075547
> calc_MSE(eva_part, holt_forecast)
[1] 5.096308
> calc_MAPE(eva_part, holt_forecast)
[1] 1.449738
>

```

Appendix G: Forecast Evaluation and Error Metrics for Holt's Method

```

> calc_MSE(as.numeric(est_part), arres_fitted)
[1] 0.8236856
> calc_MAPE(as.numeric(est_part), arres_fitted)
[1] 1.333573
> calc_MSE(as.numeric(eva_part), arres_test_fitted)
[1] 7.438177
> calc_MAPE(as.numeric(eva_part), arres_test_fitted)
[1] 1.795274

```

Appendix H: Forecast Evaluation and Error Metrics for the Adaptive Response Rate Exponential Smoothing (ARRES) Method

```

Augmented Dickey-Fuller Test
data: est_google
Dickey-Fuller = -3.817, Lag order = 12, p-value = 0.01813
alternative hypothesis: stationary
> pp.test(est_google)

Phillips-Perron Unit Root Test
data: est_google
Dickey-Fuller Z(alpha) = -33.641, Truncation lag parameter = 8, p-value = 0.01
alternative hypothesis: stationary

warning message:
In pp.test(est_google) : p-value smaller than printed p-value
> kpss.test(est_google)

KPSS Test for Level Stationarity
data: est_google
KPSS Level = 20.605, Truncation lag parameter = 8, p-value = 0.01

warning message:
In kpss.test(est_google) : p-value smaller than printed p-value

```

Appendix I: ADF, PP test and KPSS test Result before differencing.

```

Augmented Dickey-Fuller Test

data: na.omit(diff_google)
Dickey-Fuller = -12.557, Lag order = 12, p-value = 0.01
alternative hypothesis: stationary

warning message:
In adf.test(na.omit(diff_google)) : p-value smaller than printed p-value

> pp.test(na.omit(diff_google))

Phillips-Perron Unit Root Test

data: na.omit(diff_google)
Dickey-Fuller Z(alpha) = -2169.3, Truncation lag parameter = 8, p-value = 0.01
alternative hypothesis: stationary

warning message:
In pp.test(na.omit(diff_google)) : p-value smaller than printed p-value

> kpss.test(na.omit(diff_google))

KPSS Test for Level Stationarity

data: na.omit(diff_google)
KPSS Level = 0.032092, Truncation lag parameter = 8, p-value = 0.1

warning message:
In kpss.test(na.omit(diff_google)) : p-value greater than printed p-value

>

```

Appendix J: ADF, PP and KPSS test Result after first order non seasonal differencing

```

Series: value
Model: ARIMA(1,1,0) w/ drift

Coefficients:
      ar1  constant
    -0.1234   0.0329
s.e.   0.0225   0.0177

sigma^2 estimated as 0.6081: log likelihood=-2270.42
AIC=4546.83  AICC=4546.85  BIC=4563.55
> accuracy(arima110)
# A tibble: 1 x 10
  .model      .type      ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>      <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(1, 1, 0)) Training 0.0000693 0.779 0.475 -0.0209 1.07 1.00 0.991 -0.000610
>
> eva.model1 <- arima110 %>%
>   refit(new_eva_google)
> report(eva.model1)
Series: value
Model: ARIMA(1,1,0) w/ drift

Coefficients:
      ar1  constant
    -0.1234   0.0329
s.e.   0.0225   0.0177

sigma^2 estimated as 5.11: log likelihood=-1859.18
AIC=3720.35  AICC=3720.36  BIC=3725.07
> accuracy(eva.model1)
# A tibble: 1 x 10
  .model      .type      ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>      <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(1, 1, 0)) Training 0.0626 2.26 1.66 0.0407 1.45 1.01 1.01 0.109
>

```

Appendix K: Estimation and Evaluation result of ARIMA (1,1,0)

```

Series: value
Model: ARIMA(0,1,1) w/ drift

Coefficients:
      ma1  constant
-0.1221    0.0293
s.e.   0.0222    0.0155

sigma^2 estimated as 0.6082:  log likelihood=-2270.56
AIC=4547.12  AICC=4547.14  BIC=4563.84
> accuracy(arima011)
# A tibble: 1 x 10
  .model                .type      ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>                <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(0, 1, 1)) Training 0.0000751 0.779 0.475 -0.0212 1.07 1.00 0.991 -0.00176
>
> eva.model2 <- arima011 %>%
+   refit(new_eva_google)
> report(eva.model2)
Series: value
Model: ARIMA(0,1,1) w/ drift

Coefficients:
      ma1  constant
-0.1221    0.0293
s.e.   0.0222    0.0155

sigma^2 estimated as 5.107:  log likelihood=-1858.88
AIC=3719.77  AICC=3719.77  BIC=3724.49
> accuracy(eva.model2)
# A tibble: 1 x 10
  .model                .type      ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>                <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(0, 1, 1)) Training 0.0634 2.26 1.66 0.0412 1.45 1.01 1.00 0.108
>

```

Appendix L: Estimation and Evaluation result of ARIMA (0,1,1)

```

Series: value
Model: ARIMA(1,1,1) w/ drift

Coefficients:
      ar1      ma1  constant
-0.0951 -0.0287    0.0321
s.e.   0.1647  0.1651    0.0172

sigma^2 estimated as 0.6084:  log likelihood=-2270.4
AIC=4548.8  AICC=4548.83  BIC=4571.09
> accuracy(arima111)
# A tibble: 1 x 10
  .model                .type      ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>                <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(1, 1, 1)) Training 0.00000545 0.779 0.475 -0.0212 1.07 1.00 0.991 -0.000165
>
> eva.model3 <- arima111 %>%
+   refit(new_eva_google)
> report(eva.model3)
Series: value
Model: ARIMA(1,1,1) w/ drift

Coefficients:
      ar1      ma1  constant
-0.0951 -0.0287    0.0321
s.e.   0.1647  0.1651    0.0172

sigma^2 estimated as 5.11:  log likelihood=-1859.17
AIC=3720.34  AICC=3720.35  BIC=3725.07
> accuracy(eva.model3)
# A tibble: 1 x 10
  .model                .type      ME  RMSE  MAE    MPE  MAPE  MASE  RMSSE  ACF1
  <chr>                <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(1, 1, 1)) Training 0.0627 2.26 1.66 0.0408 1.45 1.01 1.01 0.109
>

```

Appendix M: Estimation and Evaluation result of ARIMA (1,1,1)

```

> report(arima212)
Series: value
Model: ARIMA(2,1,2) w/ drift

Coefficients:
      ar1      ar2      ma1      ma2      constant
    -0.2881 -0.1168  0.1639  0.0943   0.0411
s.e.      NaN    0.1585      NaN    0.1394   0.0223

sigma^2 estimated as 0.6089: log likelihood=-2270.18
AIC=4552.37 AICc=4552.41 BIC=4585.79
> accuracy(arima212)
# A tibble: 1 x 10
  .model .type ME RMSE MAE MPE MAPE MASE RMSSE ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(2, 1, 2)) Training 0.000106 0.779 0.475 -0.0207 1.07 1.00 0.991 0.000407
>
> eva.model4 <- arima212 %>%
+   refit(new_eva_google)
> report(eva.model4)
Series: value
Model: ARIMA(2,1,2) w/ drift

Coefficients:
      ar1      ar2      ma1      ma2      constant
    -0.2881 -0.1168  0.1639  0.0943   0.0411
s.e.      NaN    0.1585      NaN    0.1394   0.0223

sigma^2 estimated as 5.119: log likelihood=-1859.86
AIC=3721.71 AICc=3721.72 BIC=3726.44
> accuracy(eva.model4)
# A tibble: 1 x 10
  .model .type ME RMSE MAE MPE MAPE MASE RMSSE ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ pdq(2, 1, 2)) Training 0.0623 2.26 1.67 0.0405 1.45 1.01 1.01 0.109
>

```

Appendix N: Estimation and Evaluation result of ARIMA (2,1,2)

```

Series: value
Model: ARIMA(3,1,2)

Coefficients:
      ar1      ar2      ar3      ma1      ma2
    -1.8187 -1.0012 -0.0408  1.7338  0.8578
s.e.      0.0493  0.0653  0.0308  0.0431  0.0322

sigma^2 estimated as 0.5876: log likelihood=-2235.86
AIC=4483.72 AICc=4483.76 BIC=4517.14
> accuracy(arima_auto)
# A tibble: 1 x 10
  .model .type ME RMSE MAE MPE MAPE MASE RMSSE ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ 0 + pdq(auto_order[1], auto_order[2], auto_order[3])) Training 0.0316 0.765 0.478 0.0658 1.08 1.01 0.974 -0.00312
>
> eva.model5 <- arima_auto %>%
+   refit(new_eva_google)
> report(eva.model5)
Series: value
Model: ARIMA(3,1,2)

Coefficients:
      ar1      ar2      ar3      ma1      ma2
    -1.8187 -1.0012 -0.0408  1.7338  0.8578
s.e.      0.0493  0.0653  0.0308  0.0431  0.0322

sigma^2 estimated as 5.215: log likelihood=-1867.75
AIC=3737.49 AICc=3737.5 BIC=3742.21
> accuracy(eva.model5)
# A tibble: 1 x 10
  .model .type ME RMSE MAE MPE MAPE MASE RMSSE ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 ARIMA(value ~ 0 + pdq(auto_order[1], auto_order[2], auto_order[3])) Training 0.0914 2.28 1.68 0.0668 1.47 1.02 1.02 0.0463
>

```

Appendix O: Estimation and evaluation results of ARIMA (3,1,2)

```

> resid_arima110 <- residuals(arima110)
> Box.test(resid_arima110$resid, lag = 44, type = "Ljung-Box")

Box-Ljung test

data: resid_arima110$resid
X-squared = 189.16, df = 44, p-value < 2.2e-16

> # Result : X-squared = 189.16, df = 44, p-value < 2.2e-16
> # Decision : Reject H0 -> residuals are NOT white noise
>
> resid_arima011 <- residuals(arima011)
> Box.test(resid_arima011$resid, lag = 44, type = "Ljung-Box")

Box-Ljung test

data: resid_arima011$resid
X-squared = 194.4, df = 44, p-value < 2.2e-16

> # Result : X-squared = 194.40, df = 44, p-value < 2.2e-16
> # Decision : Reject H0 -> residuals are NOT white noise
>
> resid_arima111 <- residuals(arima111)
> Box.test(resid_arima111$resid, lag = 44, type = "Ljung-Box")

Box-Ljung test

data: resid_arima111$resid
X-squared = 189.98, df = 44, p-value < 2.2e-16

> # Result : X-squared = 189.98, df = 44, p-value < 2.2e-16
> # Decision : Reject H0 -> residuals are NOT white noise
>
> resid_arima212 <- residuals(arima212)
> Box.test(resid_arima212$resid, lag = 44, type = "Ljung-Box")

Box-Ljung test

data: resid_arima212$resid
X-squared = 190.54, df = 44, p-value < 2.2e-16

> # Result : X-squared = 190.54, df = 44, p-value < 2.2e-16
> # Decision : Reject H0 -> residuals are NOT white noise
>
> resid_arima_auto <- residuals(arima_auto)
> Box.test(resid_arima_auto$resid, lag = 44, type = "Ljung-Box")

Box-Ljung test

data: resid_arima_auto$resid
X-squared = 79.135, df = 44, p-value = 0.0009096

> # Result : X-squared = 79.135, df = 44, p-value = 0.0009096
> # Decision : Reject H0 -> residuals are NOT white noise

```

Appendix P: Result of Box-Ljung test of all models.